

Industry Report on Big Data Technologies :

Graph Databases in the Philippine Vehicle Insurance Sector

1. Introduction

In the modern digital economy, Big Data technologies are the backbone for processing vast, complex datasets. Industries ranging from healthcare to telecommunications rely on these tools to derive actionable insights. In the Philippines, the vehicle insurance sector is undergoing a massive transformation. As more Filipinos purchase vehicles, insurers are drowning in high-volume, highly connected data. While traditional relational databases (SQL) have been the standard, they struggle to reveal the "hidden" relationships between claims, drivers, and providers—relationships that often indicate organized fraud rings or complex risk patterns.

This report explores **Graph Databases**, a Big Data technology designed to treat relationships as "first-class citizens". We will analyze how this technology can revolutionize the Philippine vehicle insurance market, specifically replacing or augmenting current tabular structures that track everything from policy issuance to commission payouts.

2. Concept Overview: Graph Databases

2.1 Architecture and Working Mechanism

Unlike traditional SQL databases that store data in rigid rows and columns, a Graph Database is a NoSQL technology designed to treat the connections between data points as equally important as the data points themselves. The architecture is primarily composed of three elements:

1. **Nodes (Vertices):** These represent individual entities. In the context of your dataset, nodes would include the **Assured**, the **Vehicle (Plate Number/CS)**, the **Subagent (SSA)**, and the **Policy**.
2. **Edges (Relationships):** These are the directed links between nodes. For example, a Driver node is connected to a Vehicle node via an OWNS edge.
3. **Properties:** These are key-value pairs stored within nodes or edges. For instance, the **Car Brand**, **Car Model**, and **Vehicle Age** are properties of the Vehicle node, while the **Net Premium** and **Issue Rate** are properties of the Policy node.



2.2 Working Mechanism: Index-Free Adjacency

The core technical advantage of graph databases is "index-free adjacency." In traditional systems, to find the relationship between an **External Partner** and a specific **Claim**, the system must perform a "JOIN" operation, searching through millions of rows in an index. In a graph, each node physically points to its neighbors. The system "walks" the graph, making query speeds constant even as the dataset grows to billions of records.

3. Industry Application: The Philippine Vehicle Insurance Context

The Philippine market is unique due to its heavy reliance on agent networks (**UM/SSA**) and complex tax/commission structures (**VAT, CWT, LGT**). A graph database transforms your existing data columns into a living ecosystem.

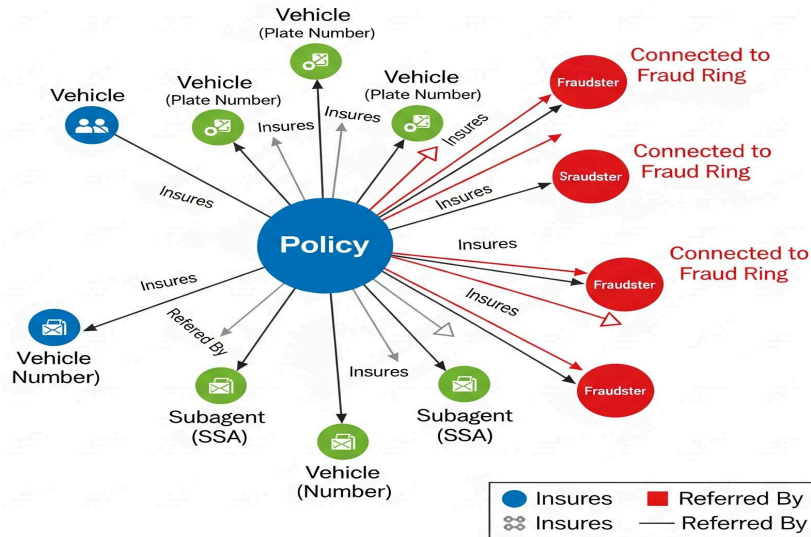
A. Fraud Ring & Link Analysis

Fraud detection is the most significant benefit of Graph Databases in the local market. Traditional spreadsheets make it nearly impossible to see if multiple "independent" claims are actually related.

- **Identify Shared Entities:** A graph can instantly show that three different **Policy Numbers** share the same **Mobile Number** or were all processed by the same **External Partner**.
- **Collision Analysis:** It can map if a specific **Plate Number** has been involved in multiple claims with different **Assured Names** over several **NAV Quarters**.

Philippine Vehicle Insurance Network Analysis: Identifying Fraudulent Relationships

Leveraging network graphs for fraud detection and claims processing



B. Revenue & Hierarchy Management

Tracking the flow of money from **Gross Premium** to **Net Revenue (Vat Exclusive)** involves multiple deductions and incentives.

- **Incentive Tracking:** By mapping the **Subagent Incentive (Payout)** and **Dealer's Mark Up** as edges, the management can see exactly how much "leakage" or profit is occurring at each step of the hierarchy.
- **Lead Source Performance:** Graphs allow for real-time visualization of which **Lead Source** or **Category** produces the highest **Gross Margin (A)** while maintaining the lowest **Cancellation Due** rate.

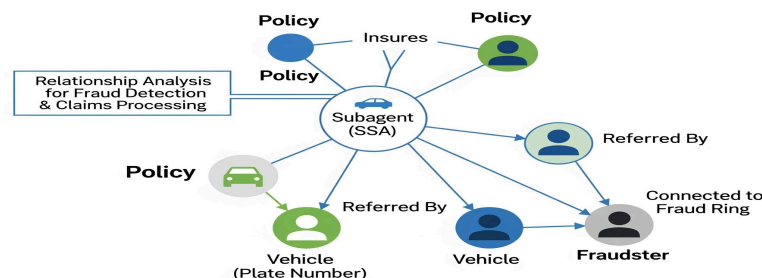
4. Industry Use Cases in the Philippines

4.1 Advanced Fraud Detection

In the Philippines, "staged accidents" often involve a network of recurring actors. Using your data points like **Mobile Number**, **Email Address**, and **Plate Number**, a graph database can perform "Pathfinding" queries. If a new applicant has a "two-degree separation" from a known fraudster (e.g., they share a common **Mortgagee** or **Lead Source**), the system can flag the application before the **Policy Issuance Date**.

4.2 Agent and Subagent (UM/SSA) Hierarchy Analysis

With fields such as **UM**, **SSA**, and **Subagent Incentive**, insurers can visualize the entire sales hierarchy. Traditionally, calculating the **Gross Profit (Vat Inclusive)** for a specific Unit Manager required aggregating thousands of rows. In a graph, one can visualize the "influence" of a UM, seeing exactly which **Lead Sources** or **Categories** they are most successful in, and identifying if their high volume is resulting in a high rate of **Cancellation Due**.



4.3 Accelerated Claims Processing

By mapping **Vehicle Age**, **Car Model**, and **History of Claims**, the graph can automate "Straight-Through Processing" (STP). If the graph shows the claimant has a clean history and the **Policy Status** is "Active" with no suspicious links to other active claims, the claim can be approved instantly, improving customer satisfaction in the competitive Manila market.

5. Benefits and Limitations

5.1 Benefits

- **Agility:** Unlike the rigid schema of your current tables, if the Philippine Insurance Commission introduces a new requirement (e.g., tracking **Road Side Assistance** usage), a graph database allows you to add this node without migrating the entire database.
- **Performance:** Relationship-heavy queries (e.g., "Which **External Partner** has the highest **Net Revenue** per **Vehicle Age**?") run 1000x faster than in SQL.
- **Data Integrity:** It ensures that a single **Plate Number** cannot be insured under two different **Assured Names** simultaneously without triggering a network alert.

5.2 Limitations

- **Skill Gap:** There is a shortage of developers in the Philippines familiar with Graph Query Languages like Cypher.
- **Storage Requirements:** Graphs require more memory (RAM) because they store all relationships in active memory to ensure high-speed traversal.

6. Future Trends

The future of the Philippine insurance industry lies in **Graph Data Science (GDS)**. By applying algorithms like "PageRank" to their data, insurers can identify the most influential **Lead Sources**. Furthermore, integrating with external data (like LTO records) via a Knowledge Graph will allow for real-time verification of **Plate Numbers** and **Engine Numbers** against national databases.

7. Conclusion

For a vehicle insurance provider handling thousands of records daily, the transition from tabular storage to Graph Databases is a strategic necessity. By focusing on the relationships between **Policy Numbers**, **Vehicles**, and **Financial Metrics**, insurers can unlock deeper insights into their **Net Revenue**, drastically reduce fraud, and gain a competitive edge in the rapidly digitizing Philippine market.

Appendix: Data Mapping Table

Current Column Name	Graph Component Type	Role in Ecosystem
Assured's Name	Node	Primary Entity
Plate Number / CS	Node	Insured Object
UM / SSA	Node	Sales Hierarchy Entity
Net Premium / VAT	Property	Attribute of the 'Policy' Node
Subagent Incentive	Property	Attribute of the 'REFERRED_BY' Edge
Policy Status	Property	State of the 'Policy' Node

